

BPL_STEM_AIR_Perfusion - test

Here we show simulations of stem cell cultivation in an aerated hollow fiber reactor. The reactor volume is kept constant and cells recycled, thus the setup is similar to perfusion cultivation.

The model combines rudimentary cell growth and metabolism combined with times series data of the metabolic rates: q_{Nmax} , q_{Lc} , and q_{O2} , marked with red in the comprehensive plot.

Ref Greuel et al: "Online measurement of oxygen enables continuous noninvasive evaluation of human-induced pluripotent stem cell (hiPSC) culture in a perfused 3D hollow-fiber bioreactor", Biotech. Bioeng., 2019.

```
In [1]: # Import the application model and context variables
        from BPL_STEM_AIR_Perfusion import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions and adapt them to the application
        from fmu_explore_pyfmi import fmu_explore

        Dummy = fmu_explore(model, parValue, parLocation, parCheck, fmu_model, fmu_proc
                             MSL_usage, MSL_version, BPL_version, \
                             opts_std, simulationTime, timeDiscreteStates, stateValue, \
                             diagrams, ax, lines)

        par = Dummy.par
        init = Dummy.init
        disp = Dummy.disp
        simu = Dummy.simu
        show = Dummy.show
        resetPen = Dummy.resetPen
        process_diagram = Dummy.process_diagram
        describe_general = Dummy.describe_general
        describe_parts = Dummy.describe_parts
        describe_MSL = Dummy.describe_MSL
        FMU_explore_info = Dummy.FMU_explore_info
        system_info = Dummy.system_info
        SDG = Dummy.SDG
```

```
In [3]: run -i BPL_STEM_AIR_Perfusion_explore.py
```

Model for the process has been setup. Key commands:

- `par()` - change of parameters and initial values
- `init()` - change initial values only
- `simu()` - simulate and plot
- `newplot()` - make a new plot
- `show()` - show plot from previous simulation
- `disp()` - display parameters and initial values from the last simulation
- `describe()` - describe culture, broth, parameters, variables with values/units

Note that both `disp()` and `describe()` takes values from the last simulation and the command `process_diagram()` brings up the main configuration

Brief information about a command by `help()`, eg `help(simu)`

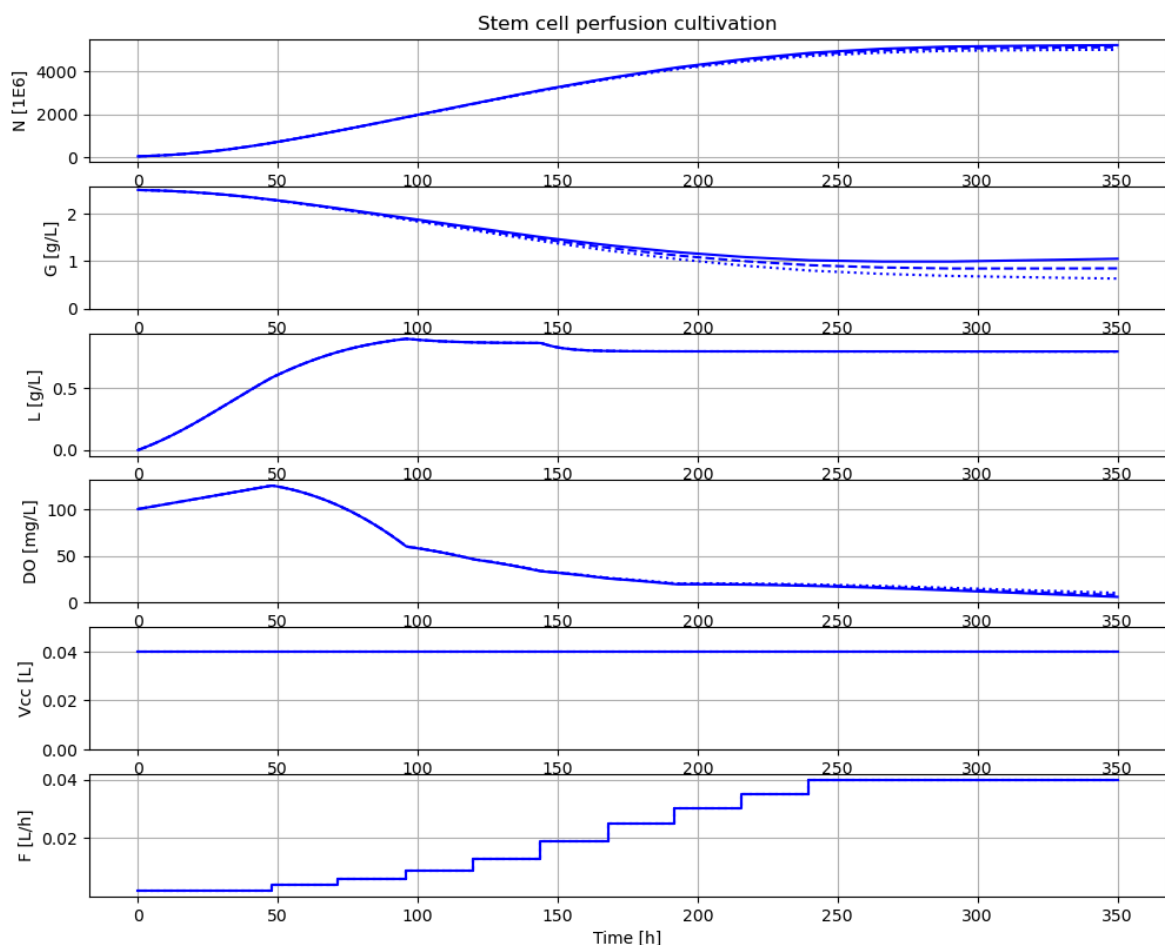
Key system information is listed with the command `system_info()`

```
In [4]: plt.rcParams['figure.figsize'] = [30/2.54, 24/2.54]
```

```
In [5]: #process_diagram()
```

```
In [6]: # Process parameters to mimic stem cell example
# - study impact of different values maintenance qm

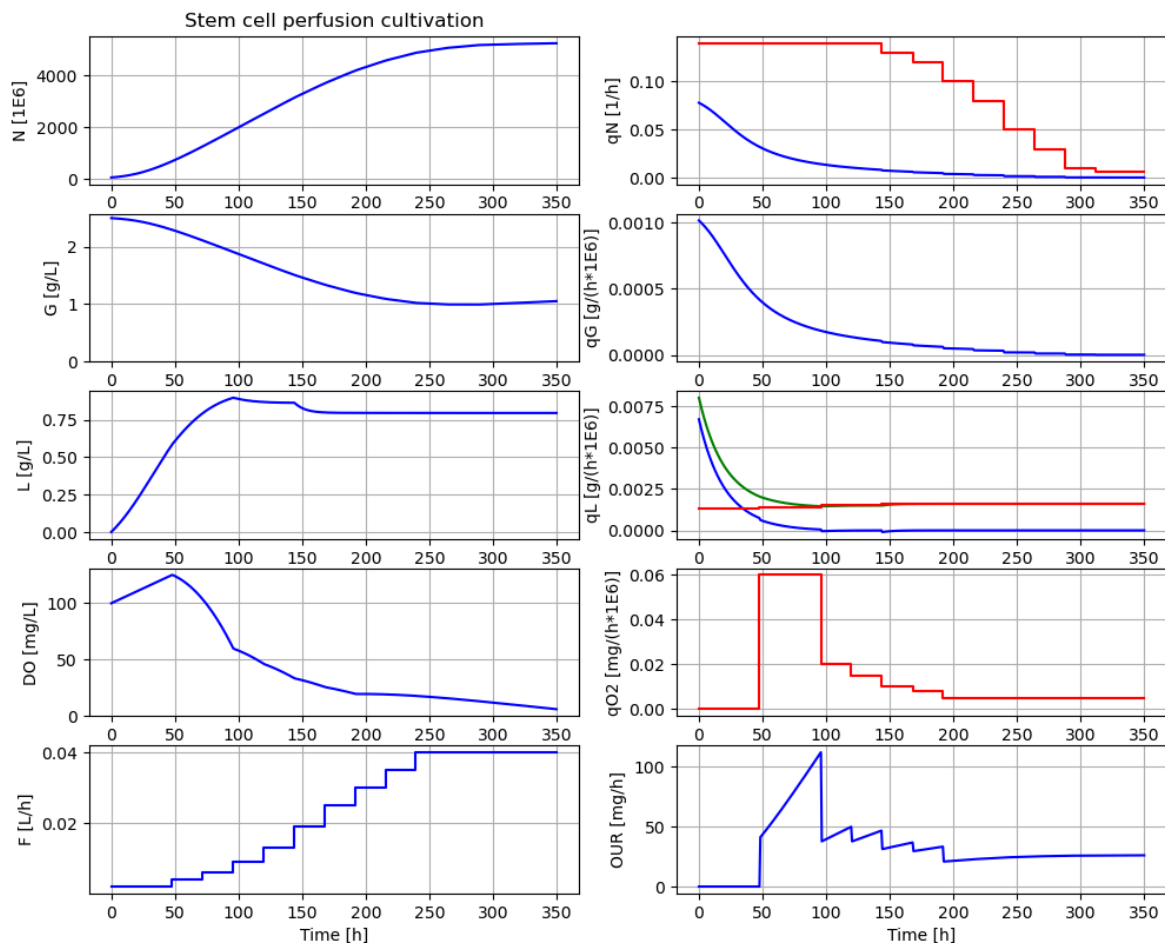
newplot(plotType='Basic')
for value in [1.0e-6, 10.0e-6, 20e-6]:
    par(qm=value)
    simu(350)
```



```
In [7]: # Process parameters to mimic stem cell example
# . comprehensvie plot including metabolic rates

par(qm=1e-6)
```

```
newplot(plotType='Comprehensive')
simu(350)
```



```
In [8]: describe('culture')
```

Reactor culture human-induced pluripotent stem cells - hiPSCs

```
In [9]: describe('Vcc')
```

Volume of the reactor : 0.04 [L]

```
In [10]: describe('N_start')
```

Number of cells at start : 50.0 [1E6]

```
In [11]: describe('N')
```

Number of cells : 5223.752 [1E6]

```
In [12]: describe('G_start')
```

Glucose conc at start : 2.5 [g/L]

```
In [13]: describe('G_in')
```

Glucose feed conc : 2.5 [g/L]

```
In [14]: describe('L')
```

Lactate conc : 0.796 [g/L]

```
In [15]: describe('DO')
```

Dissolved oxygen conc : 6.045 [mg/L]

Appendix

```
In [16]: system_info()
```

```
System information
-OS: Windows
-Python: 3.12.11
-Scipy: not installed in the notebook
-PyFMI: 2.21.0
-FMU by: JModelica.org
-FMI: 2.0
-Type: FMUModelCS2
-Name: BPL_STEM.Reactor
-Generated: 2025-07-22T09:58:37
-MSL: 4.1.0
-Description: BPL - not used
-Interaction: FMU-explore version 1.1.1
```